

ASSIGNMENT-17.1

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Task 1 – Customer Data Cleaning

Task:

Use AI to generate a Python script for cleaning a customer dataset.

Instructions:

- Handle missing values in columns (age, income, city).
- Remove duplicate records.
- Standardize text values (e.g., city names in lowercase).
- Encode categorical variables (gender, city).

Sample Code:

```
import pandas as pd  
  
data = pd.read_csv("shopping_trends.csv")  
  
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset?select=shopping_trends.csv

Expected Output:

- A cleaned Pandas data frame ready for analysis.

CODE:

```
import pandas as pd  
  
from sklearn.preprocessing import LabelEncoder  
  
# Load dataset  
  
data = pd.read_csv("/content/shopping_trends_updated.csv")
```

```

# -----
# 1. Handle Missing Values
# -----
# Age & Income → fill with median
data['Age'].fillna(data['Age'].median(), inplace=True)

data['Purchase Amount (USD)'].fillna(data['Purchase Amount (USD)'].median(),
inplace=True)


# City → fill with mode
data['Location'].fillna(data['Location'].mode()[0], inplace=True)


# -----
# 2. Remove Duplicates
# -----
data.drop_duplicates(inplace=True)


# -----
# 3. Standardize Text
# -----
data['Location'] = data['Location'].str.lower()


# -----
# 4. Encode Categorical Variables
# -----
le = LabelEncoder()

data['Gender'] = le.fit_transform(data['Gender'])
data['Location'] = le.fit_transform(data['Location'])

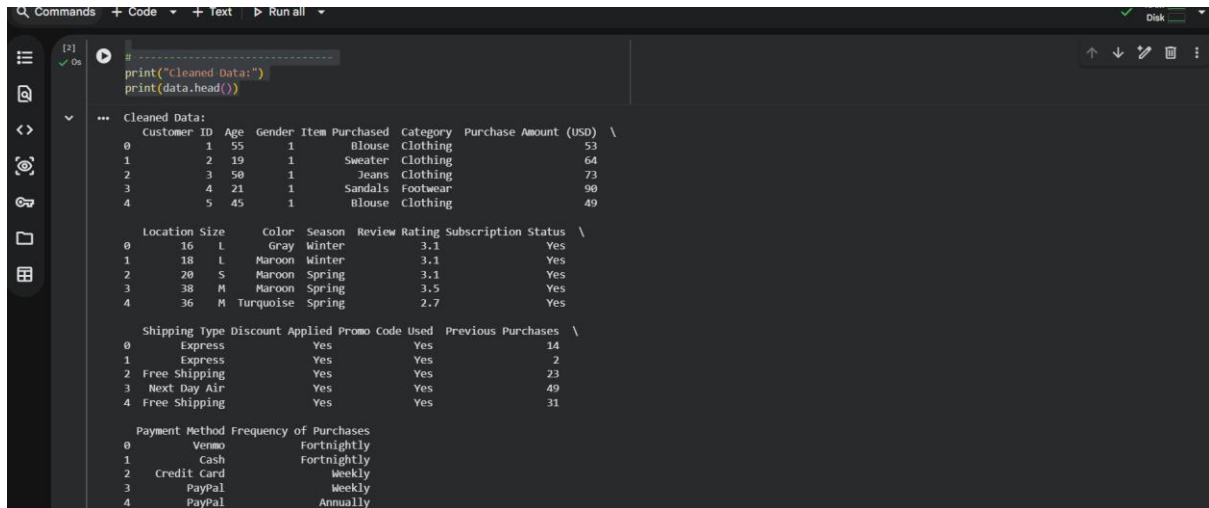

# -----

```

```
print("Cleaned Data:")
```

```
print(data.head())
```

OUTPUT:



```
[2] ✓ Os
# -----
print("Cleaned Data:")
print(data.head())
```

Cleaned Data:

	Customer ID	Age	Gender	Item	Purchased	Category	Purchase Amount (USD)
0	1	55	1	Blouse	Clothing	53	
1	2	19	1	Sweater	Clothing	64	
2	3	50	1	Jeans	Clothing	73	
3	4	21	1	Sandals	Footwear	90	
4	5	45	1	Blouse	Clothing	49	

Location Data:

	Location	Size	Color	Season	Review	Rating	Subscription Status
0	16	L	Gray	Winter	3.1	Yes	
1	18	L	Maroon	Winter	3.1	Yes	
2	20	S	Maroon	Spring	3.1	Yes	
3	38	M	Maroon	Spring	3.5	Yes	
4	36	M	Turquoise	Spring	2.7	Yes	

Shipping Data:

	Shipping Type	Discount	Applied	Promo Code	Used	Previous Purchases
0	Express	Yes	Yes	Yes	14	
1	Express	Yes	Yes	Yes	2	
2	Free Shipping	Yes	Yes	Yes	23	
3	Next Day Air	Yes	Yes	Yes	49	
4	Free Shipping	Yes	Yes	Yes	31	

Payment Method Frequency of Purchases

	Payment Method	Frequency of Purchases
0	Venmo	Fortnightly
1	Cash	Fortnightly
2	Credit Card	Weekly
3	PayPal	Weekly
4	PayPal	Annually

Task 2 – Sales Data Preprocessing

Task:

Preprocess sales transaction data using AI assistance.

Instructions:

- Convert date columns into datetime format.
- Create new features (month, year, day of week).
- Normalize sales amount column.
- Detect and handle outliers in dataset.

Sample Code:

```
import pandas as pd
```

```
data = pd.read_csv("sales_data.csv")
```

```
print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/vinothkannaeece/sales-dataset?select=sales_data.csv

Expected Output:

- A transformed dataset with time-based features and

normalized values

CODE:

```
import pandas as pd
```

```
from sklearn.preprocessing import MinMaxScaler
```

```
data = pd.read_csv("/content/sales_data.csv")
```

```
# Clean column names
```

```
data.columns = data.columns.str.strip()
```

```
print("Columns:", data.columns)
```

```
# Replace 'Date' with your actual column name
```

```
date_col = 'Sale_Date' # <-- change if needed
```

```
# Convert to datetime
```

```
data[date_col] = pd.to_datetime(data[date_col])
```

```
# Feature extraction
```

```
data['Year'] = data[date_col].dt.year
```

```
data['Month'] = data[date_col].dt.month
```

```
data['DayOfWeek'] = data[date_col].dt.day_name()
```

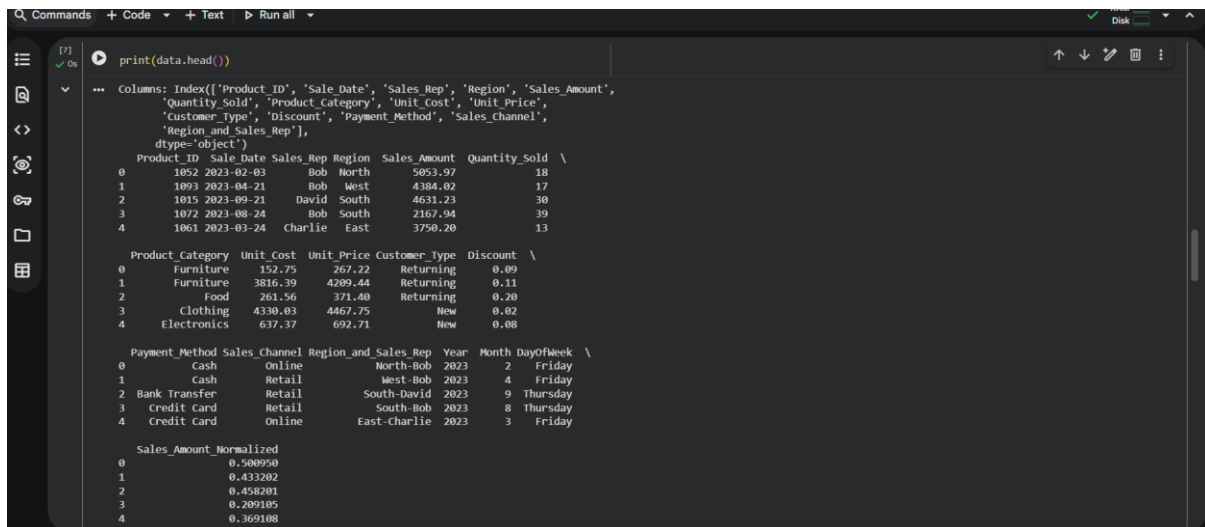
```
# Normalize (change column if needed)
```

```
scaler = MinMaxScaler()
```

```
data['Sales_Amount_Normalized'] = scaler.fit_transform(data[['Sales_Amount']])
```

```
print(data.head())
```

OUTPUT:



```
[7] ✓ Os
print(data.head())

Columns: Index(['Product ID', 'Sale Date', 'Sales_Rep', 'Region', 'Sales_Amount',
               'Quantity_Sold', 'Product_Category', 'Unit_Cost', 'Unit_Price',
               'Customer_Type', 'Discount', 'Payment_Method', 'Sales_Channel',
               'Region_and_Sales_Rep'],
              dtype='object')

Product_ID  Sale_Date Sales_Rep Region  Sales_Amount  Quantity_Sold \
0          1052 2023-02-03      Bob  North      5053.97           18
1          1091 2023-04-21      Bob  West      4384.02           17
2          1015 2023-09-21    David  South      4631.23           30
3          1072 2023-08-24      Bob  South      2167.94           39
4          1061 2023-03-24    Charlie  East      3750.20           13

Product_Category  Unit_Cost  Unit_Price  Customer_Type  Discount \
0      Furniture      152.75      267.22      Returning      0.09
1      Furniture     3816.39     4289.44      Returning      0.11
2      Food          261.56      371.40      Returning      0.20
3      Clothing     4330.03     4467.75        New          0.02
4      Electronics    637.37      692.71        New          0.08

Payment_Method  Sales_Channel  Region_and_Sales_Rep  Year  Month  DayOfWeek \
0      Cash      Online      North-Bob      2023      2      Friday
1      Cash      Retail      West-Bob      2023      4      Friday
2  Bank Transfer      Retail    South-David      2023      9      Thursday
3  Credit Card      Retail    South-Bob      2023      8      Thursday
4  Credit Card      Online      East-Charlie      2023      3      Friday

Sales_Amount_Normalized
0      0.500950
1      0.433202
2      0.458201
3      0.299105
4      0.369108
```

Task 3 – Healthcare Data Preparation

Task:

Clean and preprocess a healthcare dataset using AI scripts.

Instructions:

- Handle missing values in blood_pressure, cholesterol.
- Encode categorical features
- Scale numerical features (min-max or standard scaling).
- Split dataset into training and testing sets.

Dataset link :

<https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset>

Expected Output:

- A preprocessed dataset ready for machine learning models.

CODE:

```
import pandas as pd
```

```
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```
from sklearn.model_selection import train_test_split
```

```

# Load dataset

data = pd.read_csv("/content/dirty_v3_path.csv")

# -----

# 1. Handle Missing Values
# -----

data['Blood Pressure'] = data['Blood Pressure'].fillna(data['Blood Pressure'].mean())
data['Cholesterol'] = data['Cholesterol'].fillna(data['Cholesterol'].mean())

# -----

# 2. Encode Categorical Columns
# -----

le = LabelEncoder()
for col in data.select_dtypes(include='object').columns:
    data[col] = le.fit_transform(data[col])

# -----

# 3. Scale Numerical Features
# -----

scaler = StandardScaler()
num_cols = data.select_dtypes(include=['int64', 'float64']).columns
data[num_cols] = scaler.fit_transform(data[num_cols])

# -----

# 4. Train-Test Split
# -----

# Print columns to help identify the target variable
print("DataFrame Columns:", data.columns)

```

```
# !!! IMPORTANT: Replace 'YOUR_TARGET_COLUMN' with the actual name of your target column !!!
```

```
# Example: If 'Blood Pressure' is your target:
```

```
# target_column_name = 'Blood Pressure'
```

```
# Otherwise, choose from the printed DataFrame Columns list.
```

```
# It's also good practice to drop any columns that are not features or the target itself,
```

```
# such as 'random_notes' or 'noise_col' if they are not used for modeling.
```

```
# X = data.drop(['YOUR_TARGET_COLUMN', 'random_notes', 'noise_col'], axis=1)
```

```
# y = data['YOUR_TARGET_COLUMN']
```

```
# For now, let's assume 'HbA1c' is the target for demonstration.
```

```
# Please update 'HbA1c' to your actual target column.
```

```
target_column_name = 'HbA1c'
```

```
X = data.drop(columns=[target_column_name, 'random_notes', 'noise_col'], axis=1)
```

```
y = data[target_column_name]
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

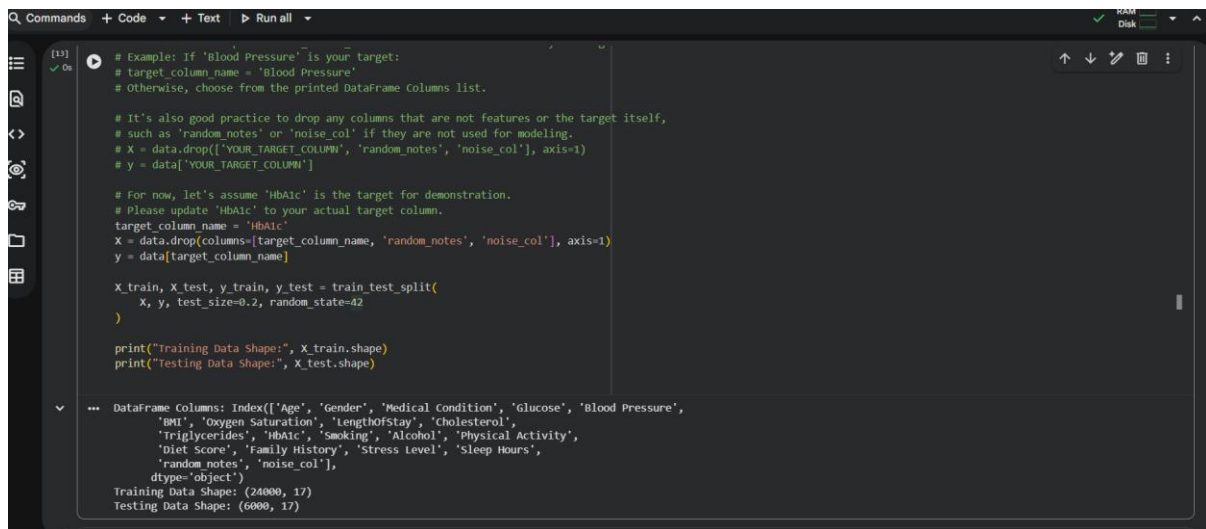
```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
print("Training Data Shape:", X_train.shape)
```

```
print("Testing Data Shape:", X_test.shape)
```

OUTPUT:



```
# Example: If 'Blood Pressure' is your target:
# target_column_name = 'Blood Pressure'
# Otherwise, choose from the printed DataFrame Columns list.

# It's also good practice to drop any columns that are not features or the target itself,
# such as 'random_notes' or 'noise_col' if they are not used for modeling.
# X = data.drop(['YOUR_TARGET_COLUMN', 'random_notes', 'noise_col'], axis=1)
# y = data['YOUR_TARGET_COLUMN']

# For now, let's assume 'HbA1c' is the target for demonstration.
# Please update 'HbA1c' to your actual target column.
target_column_name = 'HbA1c'
X = data.drop(columns=[target_column_name, 'random_notes', 'noise_col'], axis=1)
y = data[target_column_name]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)
```

--- DataFrame Columns: Index(['Age', 'Gender', 'Medical Condition', 'Glucose', 'Blood Pressure', 'BMI', 'Oxygen Saturation', 'LengthofStay', 'Cholesterol', 'Triglycerides', 'HbA1c', 'Smoking', 'Alcohol', 'Physical Activity', 'Diet Score', 'Family History', 'Stress Level', 'Sleep Hours', 'random_notes', 'noise_col'], dtype='object')

Training Data Shape: (24000, 17)
Testing Data Shape: (6000, 17)

Task 4 – Employee Salary Data Preprocessing

Task:

Clean and preprocess an employee salary dataset using AI scripts.

Instructions:

- Handle missing values
- Detect and remove salary outliers.
- Standardize department names (lowercase).
- Apply label encoding to job location, department.

Dataset link:

<https://www.kaggle.com/code/vishantmathur/employee-salary>

Expected Output:

- A structured dataset ready for HR analytics

CODE:

```
import pandas as pd
```

```
from sklearn.preprocessing import LabelEncoder
```

```
# Load dataset
```

```
data = pd.read_csv("/content/Employee_Salary_Dataset.csv")
```

```

# Print columns to verify
print("DataFrame Columns:", data.columns)

# -----
# 1. Handle Missing Values
# -----
data.fillna(method='ffill', inplace=True)

# -----
# 2. Remove Salary Outliers
# -----
Q1 = data['Salary'].quantile(0.25)
Q3 = data['Salary'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

data = data[(data['Salary'] >= lower) & (data['Salary'] <= upper)]

# -----
# 3. Standardize Text
# -----
# Removed operations for 'Department' as the column does not exist.

# -----
# 4. Encode Categorical Variables
# -----

```

```
le = LabelEncoder()
```

```
# Removed operations for 'Department' and 'Job_Location' as the columns do not exist.
```

```
# If there are other categorical columns, add them here. For example:
```

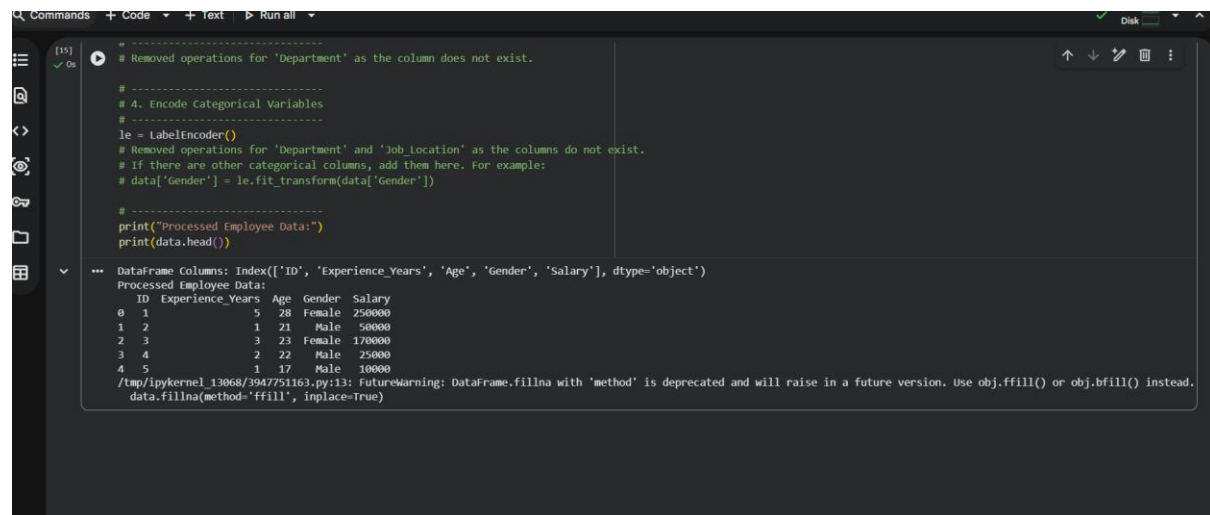
```
# data['Gender'] = le.fit_transform(data['Gender'])
```

```
# -----
```

```
print("Processed Employee Data:")
```

```
print(data.head())
```

OUTPUT:



The screenshot shows a Jupyter Notebook interface with a dark theme. The code editor on the left contains the following Python code:

```
# -----  
# Removed operations for 'Department' as the column does not exist.  
  
# 4. Encode Categorical Variables  
# -----  
le = LabelEncoder()  
# Removed operations for 'Department' and 'Job_Location' as the columns do not exist.  
# If there are other categorical columns, add them here. For example:  
# data['Gender'] = le.fit_transform(data['Gender'])  
  
# -----  
print("Processed Employee Data:")  
print(data.head())
```

The output area on the right shows the result of the code execution:

```
DataFrame Columns: Index(['ID', 'Experience_Years', 'Age', 'Gender', 'Salary'], dtype='object')  
Processed Employee Data:  
   ID  Experience_Years  Age  Gender  Salary  
0    1                 5   28  Female  250000  
1    2                 1   21   Male    50000  
2    3                 3   23  Female  170000  
3    4                 2   22   Male    25000  
4    5                 1   17   Male    10000
```

At the bottom of the output area, there is a warning message:

```
/tmp/ipykernel_13068/3947751163.py:13: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.  
data.fillna(method='ffill', inplace=True)
```